

Name: G. Venkata Varun

Rollno: 2420030187

Adaptive Software Engineering – S6

Experiment 7: Configure a local Kubernetes cluster using Minikube and deploy an Nginx Pod using a YAML file.

Kubectl: is command line tool for Kubernetes where we can directly interact with Kubernetes

Minikube: is a local Kubernetes cluster

Before that start the docker desktop engine

Step 1: Minikube installation

Go to browser type minikube installation

<https://minikube.sigs.k8s.io/docs/start/?arch=%2Fwindows%2Fx86-64%2Fstable%2F.exe+download>

To install the latest minikube **stable** release on **x86-64 Windows** using **.exe download**:

1. Download and run the installer for the [latest release](#).

Or if using PowerShell, use this command:

Click on latest release

Cmd->run as administrator

>minikube

>minikube start --driver=docker

>minikube status

If you get the error for the above commands

minikube stop

minikube delete

minikube start --driver=docker

OR

minikube delete

docker version

docker info

minikube start --driver=docker

Step 2: Installation of Kubectl

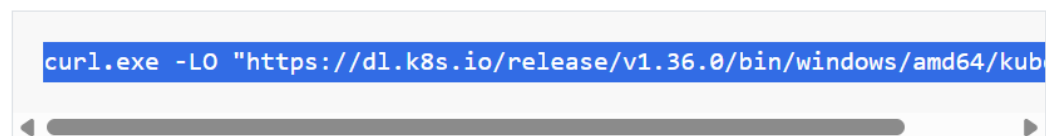
Type kubectl installation-->click on install tools-->choose platform

<https://kubernetes.io/docs/tasks/tools/install-kubectl-windows/>

Copy the Curl and paste in the cmd

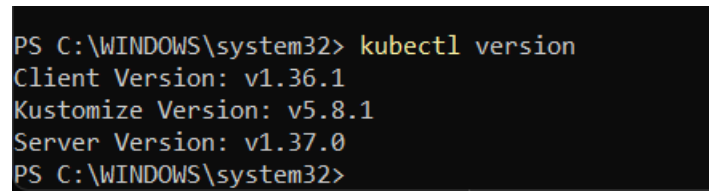
- Using curl:

If you have `curl` installed, use this command:



```
curl.exe -LO "https://dl.k8s.io/release/v1.36.0/bin/windows/amd64/kub
```

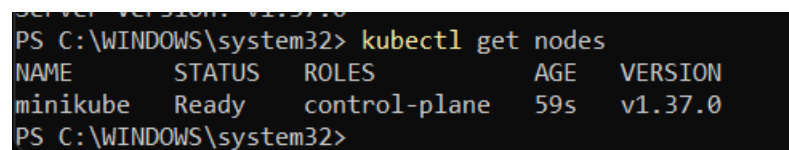
>kubectl version



```
PS C:\WINDOWS\system32> kubectl version
Client Version: v1.36.1
Kustomize Version: v5.8.1
Server Version: v1.37.0
PS C:\WINDOWS\system32>
```

Check whether kubectl is connected with minikube by the command

>kubectl get nodes



```
SERVER VERSION: v1.37.0
PS C:\WINDOWS\system32> kubectl get nodes
NAME          STATUS    ROLES          AGE    VERSION
minikube      Ready     control-plane   59s    v1.37.0
PS C:\WINDOWS\system32>
```

Step 3: Installation of POD

Goto web browser and type Kubernetes pod

<https://kubernetes.io/docs/concepts/workloads/pods/>

Pod is basically a Yaml file

Copy the yaml code

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx
spec:
  containers:
  - name: nginx
    image: nginx:1.14.2
    ports:
    - containerPort: 80
```

Cmd>vi pod.yaml

Paste the code (if vi is not recognized)

Cmd> notepad pod.yaml

Click yes

Copy the code and Save the file and close Notepad.

Deploy it:

Cmd> kubectl apply -f pod.yaml

```
PS C:\Users\vgami\Desktop> kubectl apply -f pod.yaml
pod/nginx created
```

Verify the Pod

> kubectl get pods

Get Detailed Information

> kubectl describe pod nginx

Access the Pod

kubectl port-forward nginx 8080:80

Output

```
PS C:\Users\vgami\Desktop> kubectl get pods
NAME READY STATUS RESTARTS AGE
nginx 1/1 Running 0 3m15s
PS C:\Users\vgami\Desktop> kubectl port-forward nginx 8080:80
Forwarding from 127.0.0.1:8080 -> 80
Forwarding from [::1]:8080 -> 80
Handling connection for 8080
```

Open a browser and visit:

<http://localhost:8080>

